dbt Architecture

This page helps practitioners and those interested in dbt's architecture and data flow.

About dbt architecture

The dbt application has two types of components: static and dynamic. The static components are always running to serve highly available dbt functions, like the dbt web application. On the other hand, the dynamic components are created ad-hoc to handle tasks such as background jobs or requests to use the Studio IDE.

dbt is available in most regions around the world in both single tenant (AWS and Azure) and multi-tenant configurations.

dbt uses PostgreSQL for its backend, S3-compatible Object Storage systems for logs and artifacts, and a Kubernetes storage solution for creating dynamic, persistent volumes.

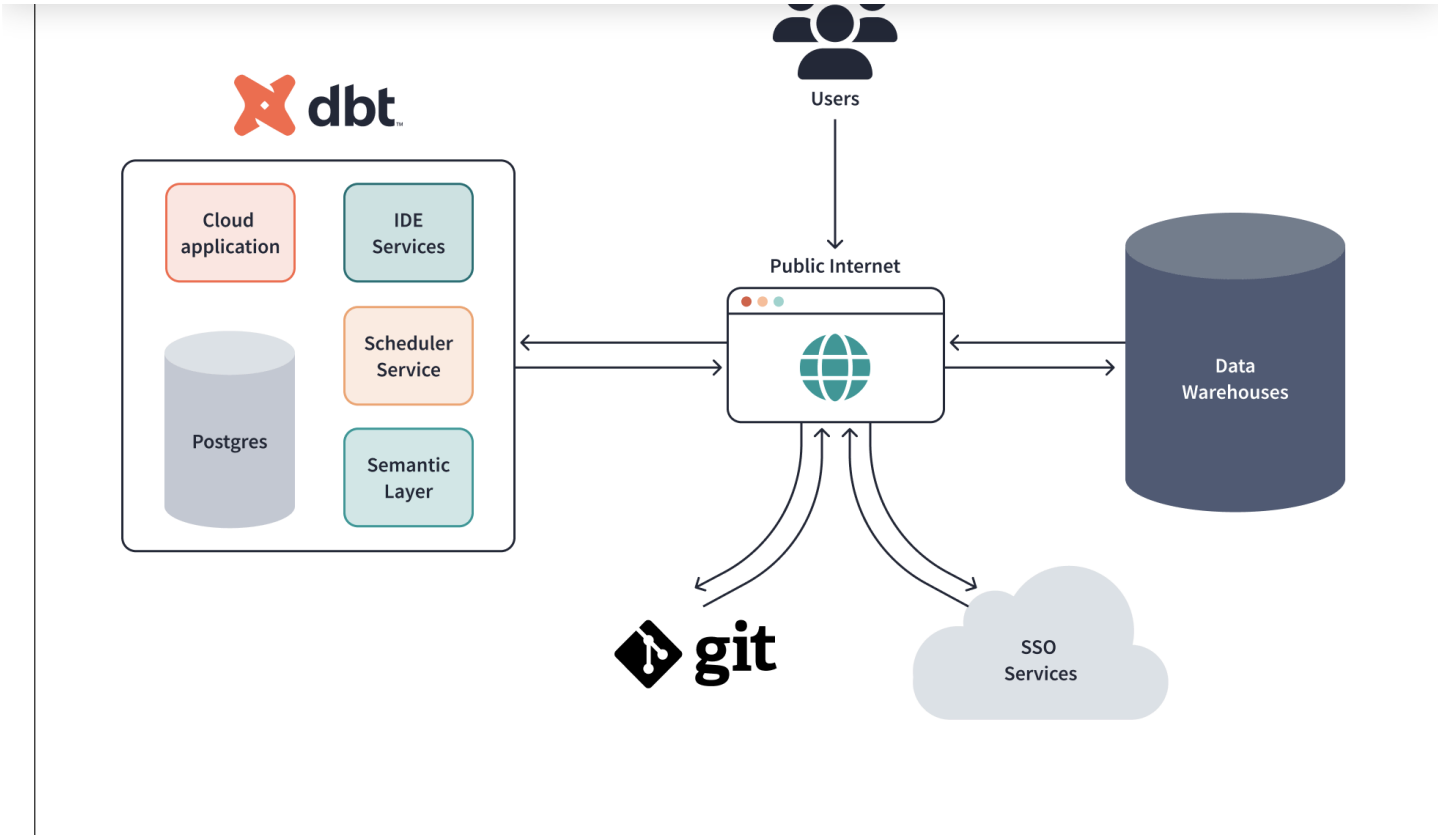
All data at rest on dbt servers is protected using AES-256 encryption.

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For a more detailed breakdown of the dbt apps, download the advanced architecture guide PDF.

Communication

dbt can communicate with several external services, including data platforms, git repositories, authentication services, and directories. All communications occur over HTTPS (attempts to connect via HTTP are redirected to HTTPS). dbt encrypts in transit using the TLS 1.2 cryptographic protocol.

TLS (Transport Layer Security) 1.2 is an industry-standard protocol for encrypting sensitive data while it travels over the public internet (which does not offer native encryption).

A typical scenario that might be seen frequently is an employee working in a public

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with TLS 1.2.

If that user runs a command that initializes communication between dbt and the data warehouse (or a git repo or an auth service) over the internet, that communication is also encrypted. This means that while the bad actor can technically see the traffic moving over that unsecured network, they can't read or otherwise parse any information. They will not be able to eavesdrop on or hack the information in any way whatsoever. They would see a nonsensical set of characters that nobody can decrypt.

For more detailed information on our security practices, read our Security page.

Data warehouse interaction

dbt's primary role is as a data processor, not a data store. The dbt application enables users to dispatch SQL to the warehouse for transformation. However, users can post SQL that returns customer data into the dbt application. This data never persists and will only exist in memory on the instance for the duration of the session. To lock down customer data correctly, proper data warehouse permissions must be applied to prevent improper access or storage of sensitive data.

Some data warehouse providers offer advanced security features that can be leveraged in dbt. Private connections allows supported data platforms on AWS to communicate with dbt without the traffic traversing the public internet. Snowflake and BigQuery offer Oauth integration which adds a layer of security for the data platforms (Enterprise-tier plan only).

Git sync

dbt can sync with a variety of git providers, including Github, Gitlab, and Azure DevOps within its integrated development environment (Studio IDE). Communication takes place over HTTPS rather than SSH and is protected using the TLS 1.2 protocol for data in transit.

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The default settings of dbt enable local users with credentials stored in dbt. Still, integrations with various authentication services are offered as an alternative, including single sign-on services. Access to features can be granted/restricted by role using RBAC.

SSO features are essential because they reduce the number of credentials a user must maintain. Users sign in once and the authentication token is shared among integrated services (such as dbt). The token expires and must be refreshed at predetermined intervals, requiring the user to go through the authentication process again. If the user is disabled in the SSO provider service, their access to dbt is disabled, and they cannot override this with local auth credentials.

Snowflake and BigQuery offer OAuth (JSON to pass info and API calls for auth) services as an alternative to SAML (XML to pass info and session cookies for auth). Users can authenticate against the data platform for secure access to dbt and prevent access when credentials are revoked.

Security

dbt Labs is dedicated to upholding industry standards for Cloud security and GDPR compliance. Our compliance certifications include the following:

- SOC2 Type II — assesses a service provider's security control environment against the trust services principles and criteria set forth by the American Institute of Certified Public Accountants (AICPA).
- ISO27001:2013 — a globally recognized standard for establishing and certifying an information security management system (ISMS).
- GDPR - dbt Labs is committed to maintaining GDPR compliance standards. Read more about our Data Processing Addendum.

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